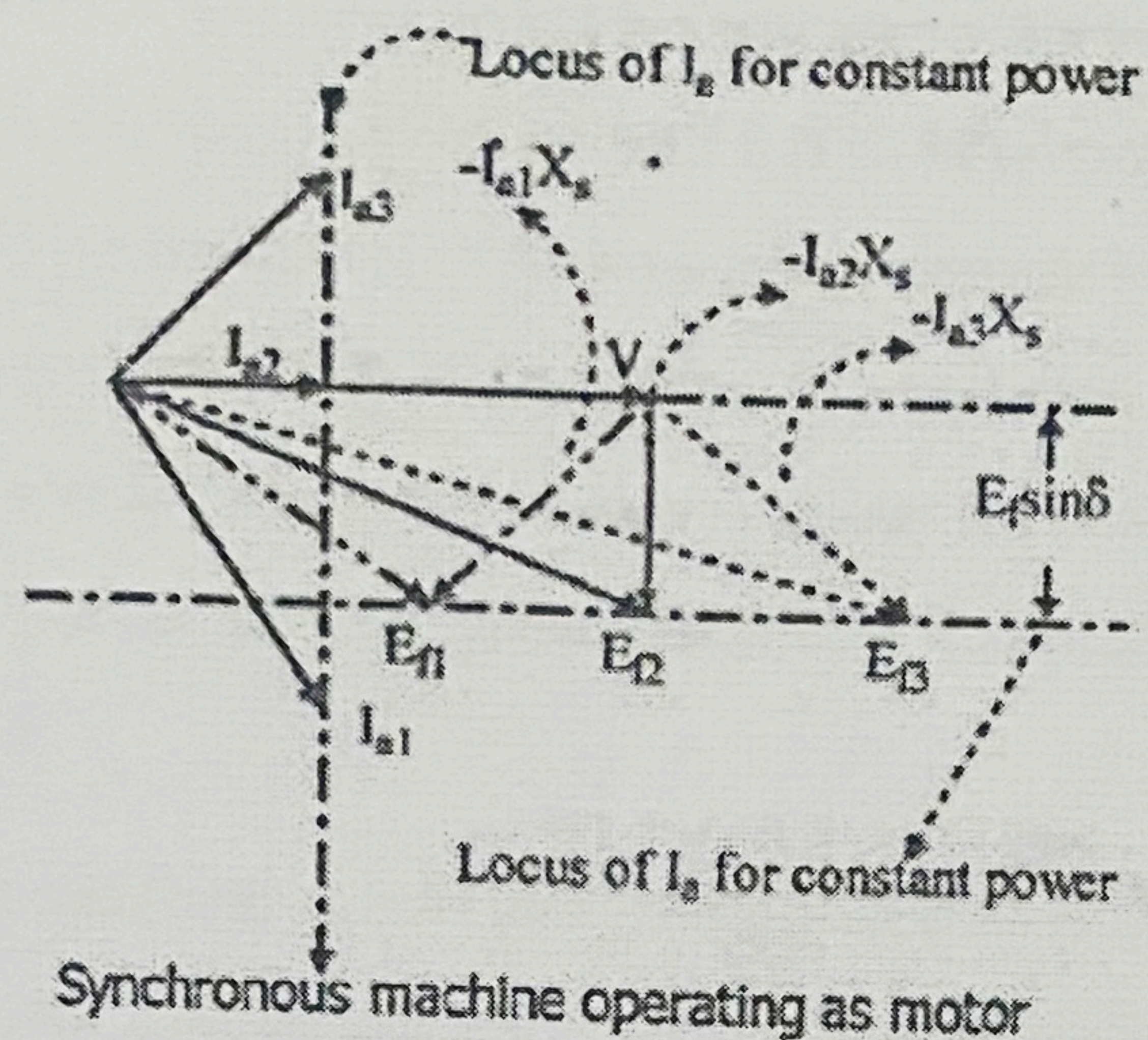
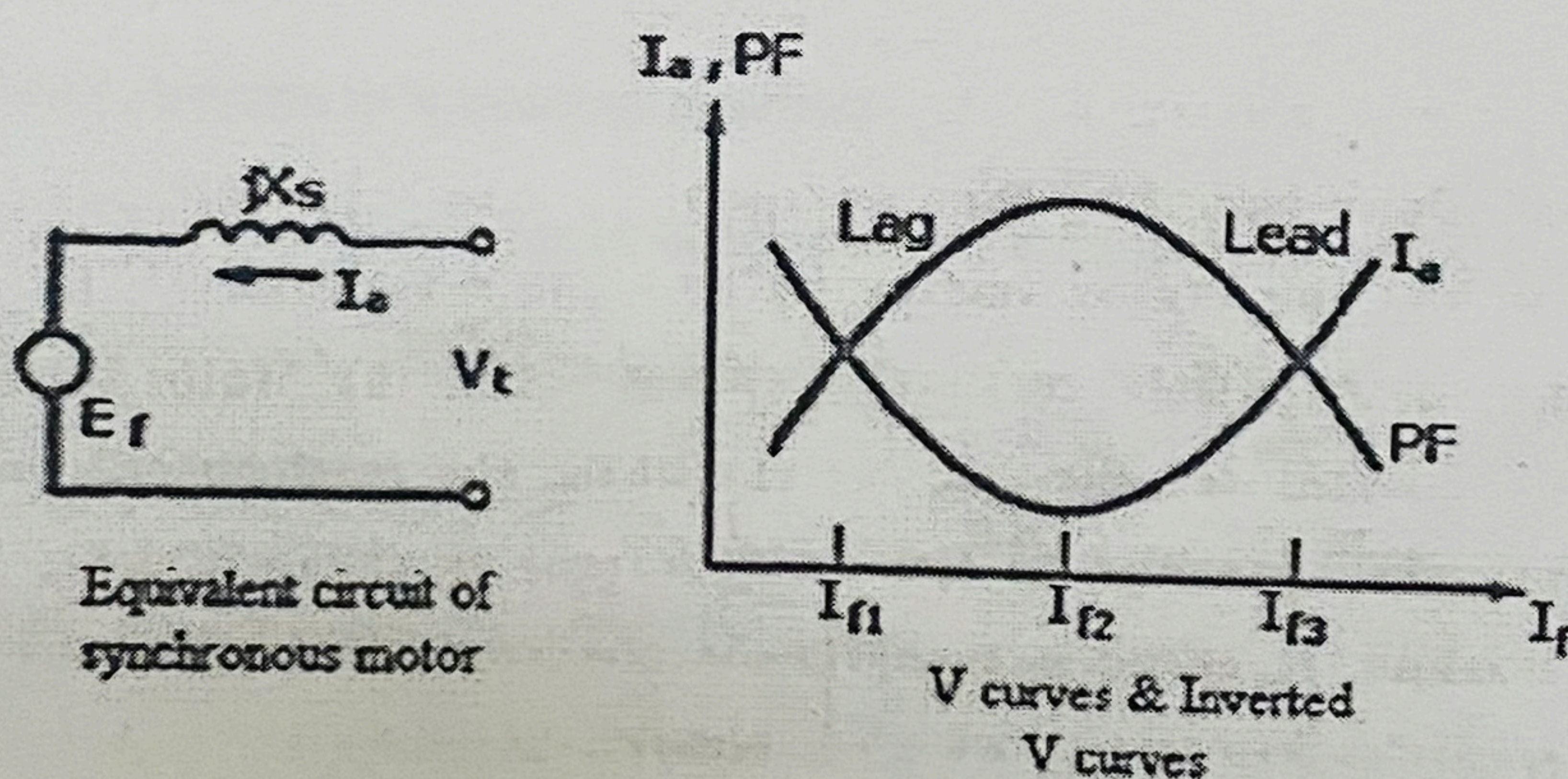




$= I_{a3}$, lagging V



For these stator currents the excitation voltages E_{f1} , E_{f2} and E_{f3} (representing the field currents I_{f1} , I_{f2} and I_{f3} respectively) are drawn to satisfy the phasor relationship $E = V - jI_a X_s$ for synchronous motor and $E = V + jI_a X_s$ for synchronous generator. The power can also be expressed as $P = 3 \frac{VE_f}{X_s} \sin \delta$. Again for constant power operation, $|E_f \sin \delta|$ is constant. Thus the locus of E_f (or I_f) is also a straight line parallel to the phasor V such that the vertical difference between the locus of E_f and the phasor V is constant and equals $E_f \sin \delta$.



The excitation voltage E_f changes linearly with the field current I_f . Therefore, as I_f is changed, E_f will change along the locus of E_f , and I_a will change along the locus of I_a , signifying a change in the power factor angle ϕ of the stator current.

When the machine is working as synchronous motor, for low field current I_{f1} , *under excitation* ($E_f = E_n$), the stator current ($I_a = I_{a1}$) is large and lagging. The stator current is minimum ($I_a = I_{a2}$) and at unity power factor for the field current I_n ($E_f = E_o$) which is called *normal excitation*. For larger field current I_n , *over excitation* ($E_f = E_G$) the stator current ($I_a = I_{a3}$) is large and leading. The variation of the stator current with the field current for constant-power operation is shown in figure. This is known as the V-curve because of the characteristic shape. The variation of the power factor with the field current is the *inverted V-curve*.